Jonathan Yu

jonathandyyu@berkeley.edu | in/jonathandyyu | (408)-335-9445

Education

University of California, Berkeley Expected May 2028

B.S. Chemistry, B.A. Computer Science; GPA: 3.9; SAT: 1580

Relevant Courses: Convex Optimization, Probability, Linear Algebra, Data Structures, Data Science, Organic Chemistry, Quantitative Chemical Analysis

Relevant Experience

Incoming Inclusion Chair, Pride Business Association *UC Berkeley–Berkeley, CA* **Spring 2026 – Present**

Catalysis Team Lead, Biofuels Technology Club *UC Berkeley–Berkeley, CA* **Fall 2024 – Present**

- Increase biofuel yields by 20% while decreasing solvent use by 50% at a laboratory scale; intended for large scale 1000L production
- Lead team of 3 undergrads to conduct experiments and coordinate with feedstock sourcing team to optimize reaction parameters

Research Experience

Researcher, Gong Lab *UC Berkeley, School of Optometry–Berkeley, CA* **January 2025 – Present**

- Co-lead first *in silico* analysis of Cx50-S50P to map structural mechanisms of gap-junction dysfunction in mouse cataract model
- Predict intrinsically disordered Periaxin variant structures *de novo* using AlphaFold3 and Rosetta3-based fragment construction
- Implement filtering, clustering, and pseudotime workflows in Seurat to identify cell-, age-, and genotype-specific cellular processes
- Analyze transcriptional dynamics in lens development in 1,000+ genes/15+ transcription factors; identified Hes1 as key regulator
- Profile differentially expressed genes with gene ontology analysis to identify candidate regulators in lens growth and homeostasis
- Maintain 500+ mice with critical genotype modifications; extracted lens tissue for proteomic and transcriptomic profiling

Peptide Design Project Lead, iGEM at Berkeley *UC Berkeley–Berkeley, CA* **January 2025 – Present**

- Identify 1NVU Ras.GTP as a target protein-protein interaction for evaluating proposed *de novo* peptide inhibitor design pipeline
- Assess 30+ candidates generated by RFDiffusion, applying ProteinMPNN sequence design and AlphaFold PAE/pLDDT scoring
- Optimize 5 partial diffusion timestep parameters to reduce backbone alignment error by 50% and increase confidence by 0.7%
- Source $2,000+ in funding and hosted 400+ attendees across 10+ industry & academic partners for Bay Area BioE Symposium

Rockefeller SSRP Intern, Fuchs Lab *The Rockefeller University – New York, NY* **June 2024 – August 2024**

- Designed and performed cell stressor/cytoprotector perturbation protocol for keratinocytes and squamous cell carcinomas
- Quantified morphological and proliferative effects using immunofluorescence imaging and segmentation pipelines in ImageJ
- Achieved 20% decrease in cell proliferation with psoriasis-associated interleukin perturbations in presence of retinoic acid
- Identified perturbation-specific transcriptional signatures in epithelial Bulk RNA-seq datasets with DEG and GSEA in R
- Presented at Rockefeller Summer Poster Session; highlighted regulatory mechanisms in skin barrier maintenance

UCSC SIP Intern, MacMillan Lab *UCSC, Department of Chemistry and Biochemistry – Santa Cruz, CA* **June 2023 – August 2023**

- Cultured and extracted novel non-enzymatic natural products from *Erythrobacter* primed with fluorinated anthranilic acid scaffold
- Developed and executed analytical methods to purify 20+ hits using iterative flash chromatography, HPLC, and 1H/19F NMR
- Interpreted LCMS fragmentation patterns to infer reactivity and structural motifs; validated structure via COSY/HSQC/HMBC NMR
- Presented at UCSC SIP Research Session, showcasing reactivity-based natural product synthesis with one final lead molecule

SSP Participant at Indiana University *The Summer Science Program – Bloomington, IL* **June 2022 – July 2022**

- Identified PdCdc14 as a key target in preventing citrus crop rot by inhibiting its mitotic phosphatase function in *P. Digitatum*
- Designed and performed enzymatic assays to quantify catalytic activity, substrate specificity, and inhibitor IC50 values
- Computationally generated homologous PdCdc14 scaffolds; verified ortholog structure conservation using sequence alignment
- Scored docking affinities of 12 FDA-approved small-molecule scaffolds and 40+ respective derivative candidates in CCG MOE
- Submitted findings on catalytic kinetics, inhibitor candidates, and active site sterics to *Purdue University Research Repository*

Skills, Interests, and Involvements

Programming & Analysis: Python, R, Java, bash scripting, HPC, biopython, Scanpy, pandas, Seurat
Modeling & Tools: Rosetta Suite/RFDiffusion, AlphaFold, PyMOL, CCG MOE, SnapGene, MNova, Xcalibur, ImageJ
Wet-Lab Techniques: NMR, LCMS, GCMS, HPLC, Protein Purification, Cell Culture, Enzyme Assays, PCR, Western Blot
Mentorship and Teaching: Berkeley CS Mentors Junior Mentor for Data C88C; SSP Connect mentor for SSP Alum
Interests: bioML, traveling, transportation networks, skiing, fragrance, floriculture, scrapbooking, zinemaking, board games